Metabolomic Data Analysis with MetaboAnalyst 6.0

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1 Background

The Pathway Analysis module combines results from powerful pathway enrichment analysis with pathway topology analysis to help researchers identify the most relevant pathways involved in the conditions under study.

There are many commercial pathway analysis software tools such as Pathway Studio, MetaCore, or Ingenuity Pathway Analysis (IPA), etc. Compared to these commercial tools, the pathway analysis module was specifically developed for metabolomics studies. It uses high-quality KEGG metabolic pathways as the backend knowledgebase. This module integrates many well-established (i.e. univariate analysis, over-representation analysis) methods, as well as novel algorithms and concepts (i.e. Global Test, GlobalAncova, network topology analysis) into pathway analysis. Another feature is a Google-Map style interactive visualization system to deliver the analysis results in an intuitive manner.

2 Data Input

The Pathway Analysis module accepts either a list of compound labels (common names, HMDB IDs or KEGG IDs) with one compound per row, or a compound concentration table with samples in rows and compounds in columns. The second column must be phenotype labels (binary, multi-group, or continuous). The table is uploaded as comma separated values (.csv).

3 Compound Name Matching

The first step is to standardize the compound labels used in user uploaded data. This is a necessary step since these compounds will be subsequently compared with compounds contained in the pathway library. There are three outcomes from the step - exact match, approximate match (for common names only), and no match. Users should click the textbfView button from the approximate matched results to manually select the correct one. Compounds without match will be excluded from the subsequently pathway analysis.

Table 1 shows the conversion results. Note: *1* indicates exact match, *2* indicates approximate match, and *0* indicates no match. A text file contain the result can be found the downloaded file *name_map.csv*

Table 1: Result from Compound

	Query	Match	HMDB	PubChem	KEGG	S
1	L-Isoleucine	Isoleucine	HMDB0000172	6306	C00407	C
2	L-Valine	L-Valine	HMDB0000883	6287	C00183	C
3	DL-Phenylalanine	D-Phenylalanine	HMDB0250791		C02265	N
4	Serine	Serine	HMDB0000187	5951	C00065	N
5	Tyrosine	L-Tyrosine	HMDB0000158	6057	C00082	N
6	3-Pyridylcarbinol	NA	NA	NA	NA	N
7	Salsoline	(R)-N-Methylsalsolinol	HMDB0003626	12989303		C
8	Urea	Urea	HMDB0000294	1176	C00086	N
9	Uracil	Uracil	HMDB0000300	1174	C00106	C

10	Syringaldehyde	NA	NA	NA	NA	N
11	Vanillin	Vanillin	HMDB0012308	1183	C00755	C
12	N-Acetyl-D-galactosamine, (R,R,R,R)	NA	NA	NA	NA	N
13	beta-Alanine	beta-Alanine	HMDB0000056	239	C00099	N
14	Phenol	Phenol	HMDB0000228	996	C00146	C
15	Picolinic acid	Picolinic acid	HMDB0002243	1018	C10164	C
16	cis-4,5-Dihydroxy-o-dithiane	NA	NA	NA	NA	N
17	2-Ethylacetoacetate	NA	NA	NA	NA	N
18	D-(+)-Glucosamine	Glucosamine	HMDB0001514	439213	C00329	N
19	L-Valyl-L-leucine	Valylleucine	HMDB0029131	6993117		C
20	Pipecolic acid	Pipecolic acid	HMDB0000070	849	C00408	C
21	L-Rhamnose	Rhamnose	HMDB0000849	25310	C00507	C
22	2-Ethylhexanoic acid	2-Ethylhexanoic acid	HMDB0031230	8697		C
23	2,5-Cyclohexadiene-1,4-dione, 2,6-bis(1,1-dimethylethyl)	NA	NA	NA	NA	N
24	2,5-Di-tert-butyl-4-hydroxyphenol	NA	NA	NA	NA	N
25	2,3,4-Trihydroxybutyric acid	Threonic acid	HMDB0000943	5460407	C01620	C
26	1,7-Dimethyl-4-(1-methylethyl)cyclodecane	NA	NA	NA	NA	N
27	Monomethyl succinate	NA	NA	NA	NA	N
28	2-Allyl-1,4-dimethoxybenzene	NA	NA	NA	NA	N
29	2-Hydroxy-3-methylbutyric acid	2-Hydroxy-3-methylbutyric acid	HMDB0000407	99823		C
30	1-Tetradecanol	Tetradecanol	HMDB0011638	8209		C
31	Hydracrylic acid	Hydroxypropionic acid	HMDB0000700	68152	C01013	C
32	Cyclohexanemethanol	NA	NA	NA	NA	N
33	2,3-Butanediol	2,3-Butanediol	HMDB0003156	262	C00265	C
34	N-(2-hydroxyethyl)-succinimide	NA	NA	NA	NA	N
35	D-Glucose	D-Glucose	HMDB0000122	5793	C00031	C
36	Methylsuccinic acid	Methylsuccinic acid	HMDB0001844	6950476	C08645	C
37	Propanoic acid, 2-oxo	NA	NA	NA	NA	N
38	2-Piperidinone	2-Piperidinone	HMDB0011749	12665		C
39	4-Methyl-5-thiazoleethanol	5-(2-Hydroxyethyl)-4-methylthiazole	HMDB0032985	1136	C04294	C
40	Xylose	D-Xylose	HMDB0000098	135191	C00181	C
41	Guanidine, N,N-dimethyl	NA	NA	NA	NA	N
42	Butyl Succinate	NA	NA	NA	NA	N
43	L-5-Oxoproline	Pyroglutamic acid	HMDB0000267	7405	C01879	C
44	l-Leucyl-l-valine	Leucyl-Valine	HMDB0028942	352038		C
45	Pimelic acid	Pimelic acid	HMDB0000857	385	C02656	C
46	1-Dodecanol	Dodecanol	HMDB0011626	8193	C02277	C
47	1-Octadecanol	Octadecanol	HMDB0002350	8221	D01924	C
48	Benzoic Acid	Benzoic acid	HMDB0001870	243	C00180	C
49	2',2,3,6'-Tetramethoxychalcone	NA	NA	NA	NA	N
50	2'-Deoxyribolactone	NA	NA	NA	NA	N
51	Cyclohexanecarboxylic acid	Cyclohexanecarboxylic acid	HMDB0031342	7413	C09822	C
52	beta-D-(-)-Ribopyranose	NA	NA	NA	NA	N
53	Decanoic acid	Capric acid	HMDB0000511	2969	C01571	C
54	d-Erythrotetrofuranose	NA	NA	NA	NA	N
55	3-Deoxyhexitol	NA	NA	NA	NA	N
56	11-Hydroxy-.DELTA.-9-tetrahydrocannabinol	NA	NA	NA	NA	N
57	Dihydrouracil	Dihydrouracil	HMDB0000076	649	C00429	C
58	Inosose-2, myo	NA	NA	NA	NA	N
59	1,3-Dioxane, 2,2-dimethyl-5-hydroxy	NA	NA	NA	NA	N
60	Myristic acid	Myristic acid	HMDB0000806	11005	C06424	C
61	Butanoic acid, 3,4-dihydroxy	NA	NA	NA	NA	N
62	D-Allofuranose	NA	NA	NA	NA	N
63	Butyl caprylate	Isomorellic acid	HMDB0030734	12294		C
64	4-Hydroxy-5-(hydroxymethyl)oxolan-2-one	NA	NA	NA	NA	N
65	Octadecyl octyl ether	NA	NA	NA	NA	N
66	Tricine	NA	NA	NA	NA	N
67	D-Mannopyranose	NA	NA	NA	NA	N
68	Azetidine, 1-acetyl-2-methyl	NA	NA	NA	NA	N
69	trans-Coniferryl alchool	NA	NA	NA	NA	N
70	5-Methyluridine	Ribothymidine	HMDB0000884	445408		C

4 Pathway Analysis

In this step, users are asked to select a pathway library, as well as specify the algorithms for pathway enrichment analysis and pathway topology analysis.

4.1 Pathway Library

There are 15 pathway libraries currently supported, with a total of 1173 pathways :

- Homo sapiens (human) [80]
- Mus musculus (mouse) [82]
- Rattus norvegicus (rat) [81]
- Bos taurus (cow) [81]
- Danio rerio (zebrafish) [81]
- Drosophila melanogaster (fruit fly) [79]
- Caenorhabditis elegans (nematode) [78]
- Saccharomyces cerevisiae (yeast) [65]
- Oryza sativa japonica (Japanese rice) [83]
- Arabidopsis thaliana (thale cress) [87]
- Escherichia coli K-12 MG1655 [87]
- Bacillus subtilis [80]
- Pseudomonas putida KT2440 [89]
- Staphylococcus aureus N315 (MRSA/VSSA)[73]
- Thermotoga maritima [57]

Your selected pathway library code is **mmu** (KEGG organisms abbreviation).

4.2 Over Representation Analysis

Over-representation analysis tests if a particular group of compounds is represented more than expected by chance within the user uploaded compound list. In the context of pathway analysis, we are testing if compounds involved in a particular pathway are enriched compared to random hits. MetPA offers two of the most commonly used methods for over-representation analysis:

- Fishers'Exact test
- Hypergeometric Test

Please note, MetPA uses one-tailed Fisher's exact test which will give essentially the same result as the result calculated by the hypergeometric test.

The selected over-representation analysis method is 'Hypergeometric test'.

4.3 Pathway Topology Analysis

The structure of biological pathways represent our knowledge about the complex relationships among molecules within a cell or a living organism. However, most pathway analysis algorithms fail to take structural information into consideration when estimating which pathways are significantly changed under conditions of study. It is well-known that changes in more important positions of a network will trigger a more severe impact on the pathway than changes occurred in marginal or relatively isolated positions.

The pathway topology analysis uses two well-established node centrality measures to estimate node importance - **degree centrality** and **betweenness centrality**. Degree centrality is defined as the number of links occurred upon a node. For a directed graph there are two types of degree: in-degree for links come from other nodes, and out-degree for links initiated from the current node. Metabolic networks are directed graph. Here we only consider the out-degree for node importance measure. It is assumed that nodes upstream will have regulatory roles for the downstream nodes, not vice versa. The betweenness centrality measures the number of shortest paths going through the node. Since the metabolic network is directed, we use the relative betweenness centrality for a metabolite as the importance measure. The degree centrality measure focuses more on local connectivities, while the betweenness centrality measure focuses more on global network topology. For more detailed discussions on various graph-based methods for analyzing biological networks, please refer to the article by Tero Aittokallio, T. et al. [1]

Please note, for comparison among different pathways, the node importance values calculated from centrality measures are further normalized by the sum of the importance of the pathway. Therefore, the total/maximum importance of each pathway is 1; the importance measure of each metabolite node is actually the percentage w.r.t the total pathway importance, and the pathway impact value is the cumulative percentage from the matched metabolite nodes.

Your selected node importance measure for topological analysis is 'relative betweenness centrality'.

5 Pathway Analysis Result

The results from pathway analysis are presented graphically as well as in a detailed table.

A Google-map style interactive visualization system was implemented to facilitate data exploration. The graphical output contains three levels of view: **metabolome view**, **pathway view**, and **compound view**. Only the metabolome view is shown below. Pathway views and compound views are generated dynamically based on your interactions with the visualization system. They are available in your downloaded files.

[1]Tero Aittokallio and Benno Schwikowski. *Graph-based methods for analyzing networks in cell biology*, Briefings in Bioinformatics 2006 7(3):243-255

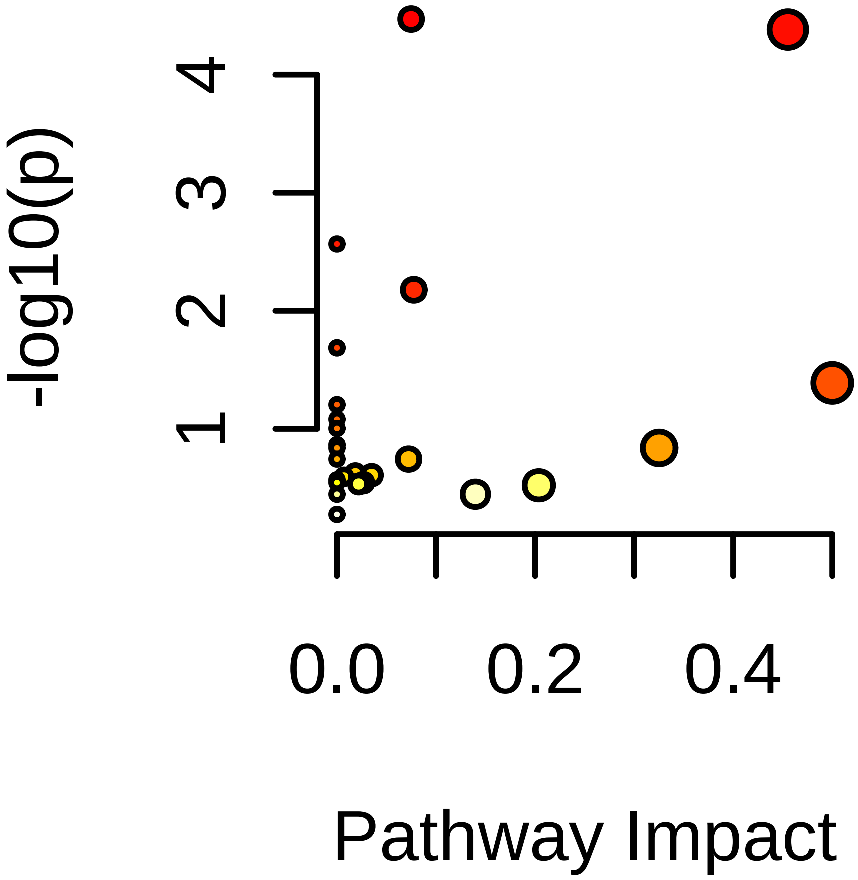

Figure 1: Summary of Pathway Analysis

The table below shows the detailed results from the pathway analysis. Since we are testing many pathways at the same time, the statistical p values from enrichment analysis are further adjusted for multiple testings. In particular, the **Total** is the total number of compounds in the pathway; the **Hits** is the actually matched number from the user uploaded data; the **Raw p** is the original p value calculated from the enrichment analysis; the **Holm p** is the p value adjusted by Holm-Bonferroni method; the **FDR p** is the p value adjusted using False Discovery Rate; the **Impact** is the pathway impact value calculated from pathway topology analysis.

Table 2: Result from Pathway Analysis

	Total	Expected	Hits	Raw p	-log10(p)	Holm adjust	FDR	Impact
Pantothenate and CoA biosynthesis	20	0.21	4	3.38E-05	4.47E+00	2.71E-03	1.66E-03	0.07
beta-Alanine metabolism	21	0.22	4	4.15E-05	4.38E+00	3.28E-03	1.66E-03	0.46
Valine, leucine and isoleucine biosynthesis	8	0.08	2	2.72E-03	2.57E+00	2.12E-01	7.25E-02	0.00
Pyrimidine metabolism	39	0.40	3	6.65E-03	2.18E+00	5.12E-01	1.33E-01	0.08
Propanoate metabolism	22	0.23	2	2.06E-02	1.69E+00	1.00E+00	2.75E-01	0.00
Neomycin, kanamycin and gentamicin biosynthesis	2	0.02	1	2.06E-02	1.69E+00	1.00E+00	2.75E-01	0.00
Phenylalanine, tyrosine and tryptophan biosynthesis	4	0.04	1	4.08E-02	1.39E+00	1.00E+00	4.67E-01	0.50
Valine, leucine and isoleucine degradation	40	0.41	2	6.25E-02	1.20E+00	1.00E+00	6.25E-01	0.00
Fatty acid biosynthesis	47	0.49	2	8.30E-02	1.08E+00	1.00E+00	7.38E-01	0.00
Phenylalanine metabolism	10	0.10	1	9.92E-02	1.00E+00	1.00E+00	7.94E-01	0.00
Arginine biosynthesis	14	0.15	1	1.36E-01	8.66E-01	1.00E+00	8.94E-01	0.00
D-Amino acid metabolism	15	0.16	1	1.45E-01	8.38E-01	1.00E+00	8.94E-01	0.00
Starch and sucrose metabolism	15	0.16	1	1.45E-01	8.38E-01	1.00E+00	8.94E-01	0.33
Ubiquinone and other terpenoid-quinone biosynthesis	19	0.20	1	1.81E-01	7.43E-01	1.00E+00	9.63E-01	0.00
Pentose and glucuronate interconversions	19	0.20	1	1.81E-01	7.43E-01	1.00E+00	9.63E-01	0.07
One carbon pool by folate	26	0.27	1	2.39E-01	6.22E-01	1.00E+00	1.00E+00	0.02
Galactose metabolism	27	0.28	1	2.47E-01	6.07E-01	1.00E+00	1.00E+00	0.03
Glutathione metabolism	28	0.29	1	2.55E-01	5.94E-01	1.00E+00	1.00E+00	0.01
Lysine degradation	30	0.31	1	2.71E-01	5.68E-01	1.00E+00	1.00E+00	0.00
Sphingolipid metabolism	32	0.33	1	2.86E-01	5.44E-01	1.00E+00	1.00E+00	0.00
Glyoxylate and dicarboxylate metabolism	32	0.33	1	2.86E-01	5.44E-01	1.00E+00	1.00E+00	0.03
Cysteine and methionine metabolism	33	0.34	1	2.93E-01	5.32E-01	1.00E+00	1.00E+00	0.02
Glycine, serine and threonine metabolism	34	0.35	1	3.01E-01	5.22E-01	1.00E+00	1.00E+00	0.20
Amino sugar and nucleotide sugar metabolism	42	0.44	1	3.58E-01	4.46E-01	1.00E+00	1.00E+00	0.00
Tyrosine metabolism	42	0.44	1	3.58E-01	4.46E-01	1.00E+00	1.00E+00	0.14
Purine metabolism	71	0.74	1	5.31E-01	2.75E-01	1.00E+00	1.00E+00	0.00

6 Appendix: R Command History

```
 [1] "mSet<-InitDataObjects(\"conc\", \"pathora\", FALSE, 150)"
 [2] "cmpd.vec<-c(\"L-Isoleucine\",\"L-Valine\",\"DL-Phenylalanine\",\"Serine\",\"Tyrosine\",\"3-Pyr
 [3] "mSet<-Setup.MapData(mSet, cmpd.vec);"
 [4] "mSet<-CrossReferencing(mSet, \"name\");"
 [5] "mSet<-CreateMappingResultTable(mSet)"
 [6] "mSet<-SetKEGG.PathLib(mSet, \"mmu\", \"current\")"
 [7] "mSet<-SetMetabolomeFilter(mSet, F);"
 [8] "mSet<-CalculateOraScore(mSet, \"rbc\", \"hyperg\")"
 [9] "mSet<-PlotPathSummary(mSet, F, \"path_view_0_\", \"png\", 150, width=NA, NA, NA )"
[10] "mSet<-SaveTransformedData(mSet)"
[11] "mSet<-PreparePDFReport(mSet, \"guest12070946642183954316\")\n"
```

The report was generated on Thu Feb 26 15:33:06 2026 with R version 4.5.2 (2025-10-31), OS system: Linux.